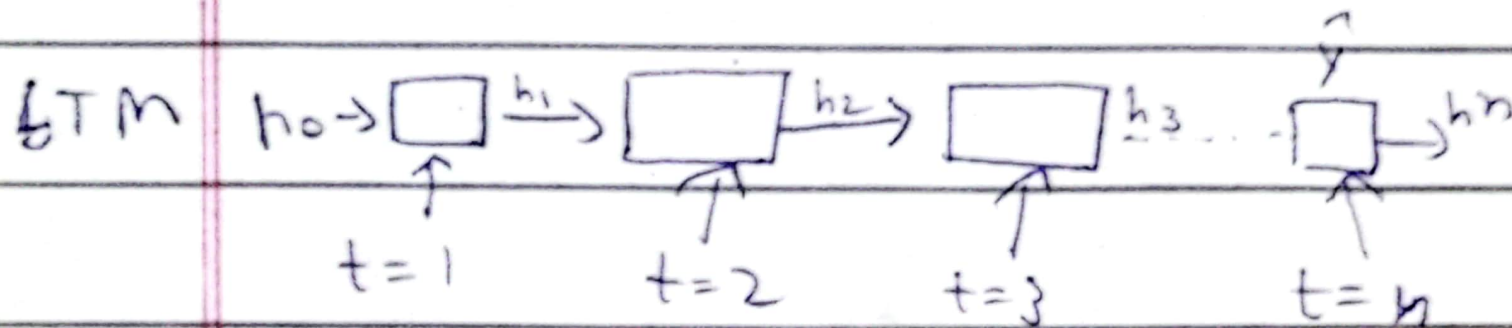


LONG short term memory:- (LSTM):-

What
part

LTM



Story of vikram.

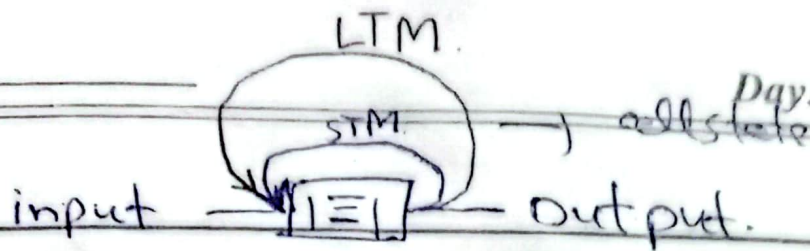
→ So in LSTM u maintain a long term memory path.

→ If u know that something is important u put it to LTM from STM.

• So there are 2 differences from RNN.

Date: _____

Day: _____

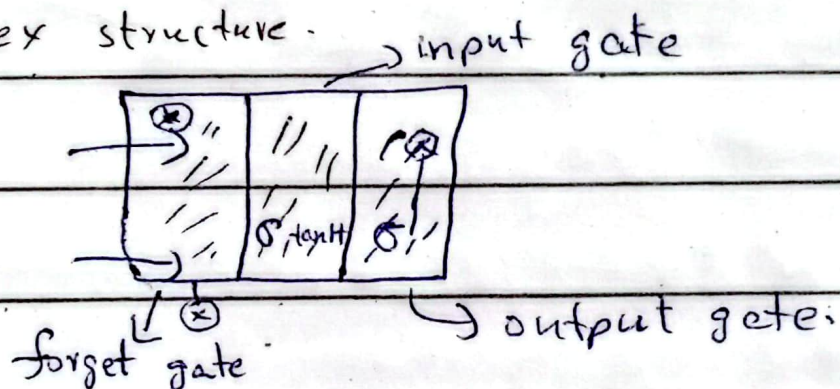


This is 1st difference.

→ 2nd the architecture inside is very complex as it to handle STM LTM & Talk blw them..

Communication blw them.

Complex structure.



forget gate:- input on this LTM → remove
basis what
to remove from
LTM.

simply decide what to be & what
to be removed from LTM.

Input gate:- LTM what → add
to be

Output gate:- make decision on LTM
& current input.

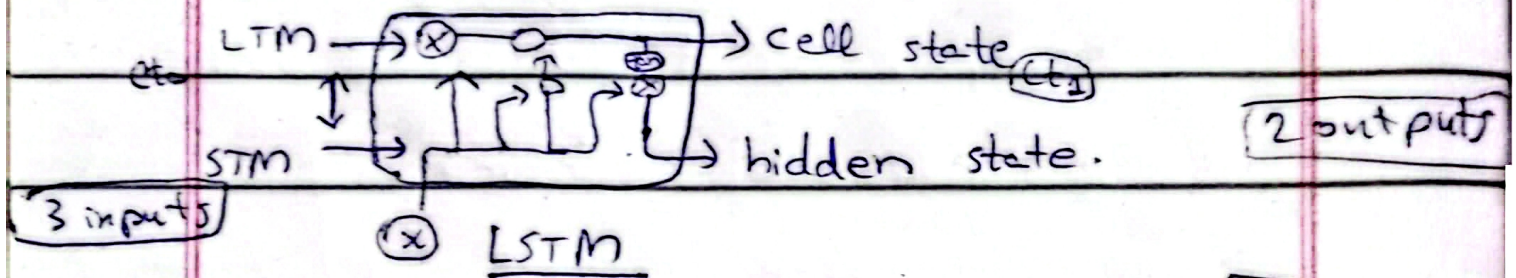
$C_t \rightarrow$ cell state
 $h_t \rightarrow$ hidden "

Date: _____

Day: _____

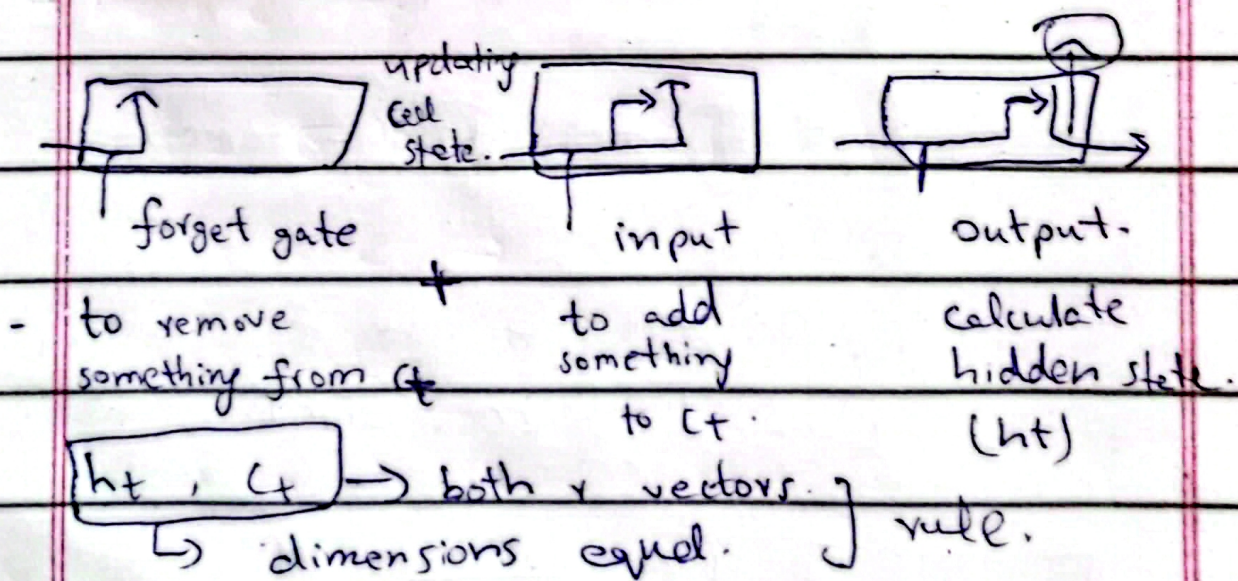
$\rightarrow$ Also create short term memory on every given stage.

LSTM [The How] :-



- $\rightarrow$ Inside update cell state $\{ \begin{matrix} \rightarrow x_t & \boxed{C_{t-1}} & \boxed{\text{remove}} \\ \rightarrow x_t & \boxed{\phantom{C_{t-1}}} & \boxed{\text{add}} \end{matrix}$
- $\rightarrow$ Calculate h_t .

So in processing 3 steps involved.



$x_t \rightarrow$ just a word converted to vectors.

Date: _____

Day: _____

$\overline{f_t}$ i_t $\overline{c_t}$ o_t
forget input gate output gate.
gate
all of them are vectors.

→ Dimensions also same exactly h_t, c_t → rule

Pointwise operations:-

$$\otimes \quad \begin{matrix} c_{t-1} & [4 & 5 & 6] \\ f_t & [1 & 2 & 3] \end{matrix} = c_{t-1} \times f_t = [4 \ 10 \ 18]$$

$$\oplus \quad c_{t-1} + f_t = [5 \ 7 \ 9]$$

$$\tanh \quad [\tanh(4) \quad \tanh 5 \quad \tanh 6] = [0.26 \ 0.34 \ 0.52]$$

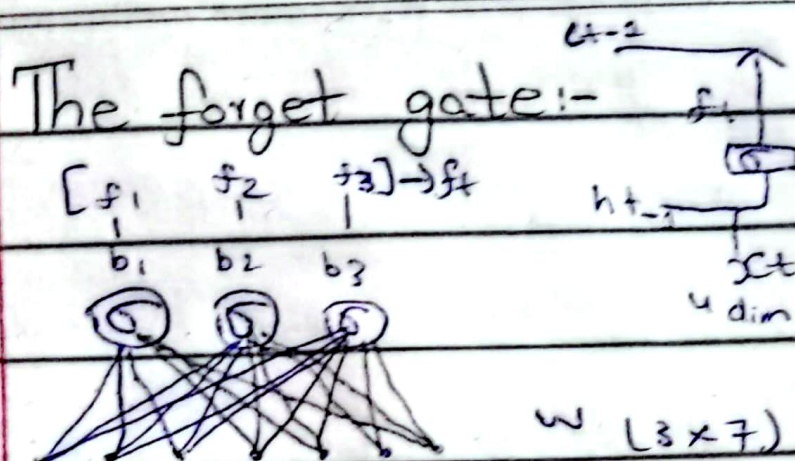
shape/dimen → same & output will be a vector.

→ Also there are nodes of a neural network & each node has act. function.

→ And how many nodes → ν decided by you as it is hyperparameter.

Date: _____

Day: _____



$w (3 \times 7) = 21$ weights

$$f_t = \sigma \left(\underbrace{w_f}_{(3 \times 7)} \underbrace{[h_{t-1}, x_t]}_{7 \times 1} + \underbrace{b_f}_{3 \times 1} \right)$$

$[3 \times 1] = [3 \times 1]$

$$f_t \otimes c_{t-1} \quad c_{t-1} \rightarrow \text{remove.}$$

The input gate:- we do filtering.

$$c_t = \tanh(W_c [h_{t-1}, x_t] + b_c)$$

↳ Potential imp information.

$$i_t = \sigma(W_i [h_{t-1}, x_t] + b_i)$$

$$i_t \otimes \bar{c}_t \rightarrow \text{new quantity.}$$

$$f_t \otimes c_{t-1} \rightarrow \text{from forget gate.}$$

$$\text{final eq.} = c_t = f_t \otimes c_{t-1} \oplus i_t \otimes \bar{c}_t$$

→ If we know something is not new important come so make it [1 1 1]

Date: _____

Day: _____

then it will go exactly same.

→ AND if there is nothing to add new $[0 \ 0 \ 0]$.

Output gate:-

$$h_t = [o_t \otimes \tanh(c_t)] \rightarrow \text{output}$$

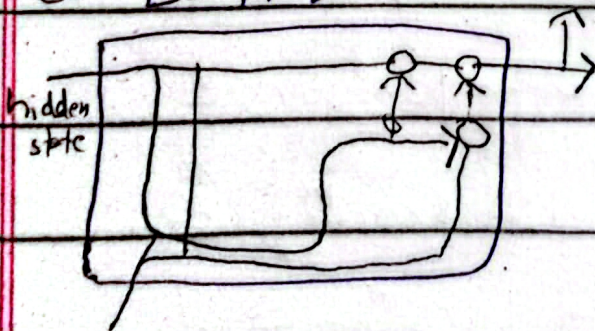
$$o_t = \sigma(w_o[h_{t-1}, x_t] + b_o)$$

Gated Recurrent unit:- (GRU)

It's an RNN architecture, used to process sequential data.

→ GRU in comparison to LSTM gives a simpler architecture. (So less no of parameter).

→ Also its performance is comparable to LSTM.



→ No concept of cell state.

→ On single state carry both long

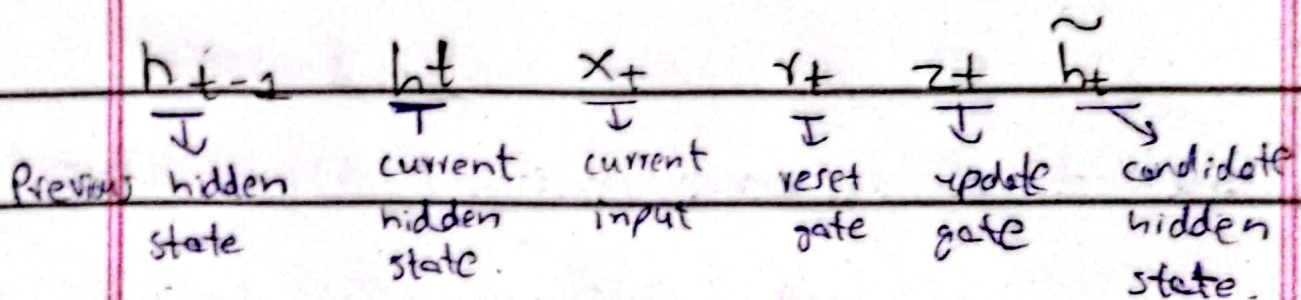
Date: _____

Day: _____

x short term context.

→ Only 2 gates not 3.
Reset gate Update gate.

• Goal is to calculate output from given input.



• All these are vectors (set of numbers)

Dimensionality are same for all except x_t .

σ $\tanh$

Neural network layers (fully connected).

→ All have same N.N layers.

→ no of nodes always same & the no u specify will exactly same for the vectors (dimensionality).

Date: _____

Day: _____

$\otimes$ $\oplus$ $\ominus$:- These are pointwise operations.

Now we have h_{t-1} , x_t $\xrightarrow{\text{calculate}}$ h_t

4 steps :- a) calculate r_t (reset gate).

b) Calculating $\tilde{h}_t$ (candidate hidden state).

c) calculate z_t (update gate).

d) calculate h_t (current hidden state).

